Target Triage with Protein Language Models: Lung Adenocarcinoma (LUAD)

BIOT 6900 · Module 3 Assignment · Bhuvanesh Karthik Nilavalagan

Background

Module 2 gave me a ranked list of 15 genes that looked suspicious across RNA, protein, and genetic evidence in lung adenocarcinoma. This assignment asks a different question about those same 15 genes. Even if a gene is disrupted in cancer, could we actually make a drug against it? I used a protein language model called ESM C to check whether each of my targets resembles proteins we already know how to drug, and whether that resemblance points toward a small molecule drug or an antibody. I had AI-assisted help writing the Python side of this, since that is still the part I am least confident with on my own, but the biological reasoning below, what the numbers mean and whether I actually believe them, is mine.

All 15 of my genes had a match in the reference protein database, so I did not lose anyone to “no data available” this round. Before training anything, I removed my own targets from the training panel so that no gene could be predicted using its own label. Only one of my genes (PALM) was in the panel, so the model was trained on the remaining 1,200 proteins.

1. Did the new evidence agree with my Module 2 ranking?

Mostly, at the top. FGFR2 was my number one gene in Module 2 and stayed number one here too. It had the strongest multi-omics evidence back then, and now it also looks like the protein in my whole list that most resembles an already successfully drugged target. That kind of agreement across two completely different methods is reassuring.

The biggest drop happens in the middle of the rankings, where BRCA2 fell six spots, from sixth to twelfth. Its Module 2 evidence was strong, showing disruption at both the RNA and protein level, but this new model found nothing in its sequence resembling any known drug target, in either modality (p_sm 0.00, p_ab 0.00). That actually makes sense biologically. BRCA2 isn’t an enzyme with a pocket to block, it’s a large structural protein that helps repair DNA, and no drug works by binding a protein like it directly, so the model has nothing to compare it to.

The biggest rise was SLC12A7, which jumped five spots from twelfth to seventh because of a high small-molecule prior (p_sm 0.85). It is a membrane ion transporter, and its nearest drugged neighbours are ion channels (KCNA2, SCN4A), a group of proteins that small molecules often target. CYP1A1 and VWF also each moved up three spots for the same reason: they look like proteins with existing drugs.

I weighted the two kinds of evidence sixty-forty in favour of Module 2 (w = 0.6), since that evidence came from real LUAD patient data across three separate biological layers. This week’s evidence only shows that a protein resembles something we’ve drugged before, which is useful but a step removed from actually being tied to lung cancer. I didn’t want a gene with weak cancer evidence to jump ahead of one with strong cancer evidence just because its protein sequence happens to look like some existing drug target’s.

2. One target, two very different drug stories: VWF and FGFR2

Two of my genes make an almost textbook-clean contrast for how a protein's physical location in the body decides what kind of drug could even reach it.

Gene	p_sm (small molecule)	p_ab (antibody)	Nearest drugged proteins (cosine similarity)
FGFR2	0.86	0.12	FGFR1 (0.99), GABRG2 (0.95), GABRG3 (0.95)
VWF	0.13	0.76	PLAT (0.96), MMP9 (0.96), TPO (0.95)

VWF is a protein that floats freely in the bloodstream, and its whole job is clotting, done entirely outside of cells. Its closest matches were mostly other proteins that work outside cells: PLAT (tissue plasminogen

activator, a blood protein) and MMP9 (a secreted enzyme). That makes sense, since antibodies are large molecules themselves and well suited to grabbing onto something big and exposed in the open. A small molecule would have nothing to grip onto or slip inside of here, since there is no enclosed pocket, just a big protein sitting in plasma. This prediction also matches the real world: VWF already has an approved antibody-type drug, caplacizumab, used for a rare clotting disorder.

FGFR2 is the opposite kind of protein entirely. It is a receptor that sits in the cell membrane, with part of it poking inside the cell, and that inside part is a kinase enzyme with an actual binding pocket, the kind of thing small molecules are specifically good at wedging into. Unsurprisingly, its closest match was FGFR1 (0.99), a nearly identical sibling receptor that already has small molecule drugs on the market.

Same assignment and same scoring method, yet two genes end up with completely different drug development paths, because the biology of where each protein physically sits determines which kind of drug even makes sense to pursue.

3. How much should I trust this scoring method?

I validated the underlying model two different ways. The easy way, just randomly splitting proteins into training and testing groups, made the model look very good. But that is a bit of a trick, because close relatives of a protein I am testing were often still sitting in the training data, so the model could partly just recognize something it had basically already seen rather than actually learning what makes a protein druggable in general.

Split	AUROC	Average precision
Random split	0.850	0.737
Family split	0.737	0.541
Chance level	0.500	0.334

The honest version holds out entire protein families at once, so the model has to judge a type of protein it has truly never encountered. Performance dropped noticeably under this harder test, which is exactly what I would expect if the easy version was partly cheating. Even so, the honest version still performed clearly better than random guessing (average precision 0.541 vs. a chance level of 0.334), so I do not think the whole thing is noise. It is just a more modest and believable level of skill than the easy test would suggest. The random split would have hidden about 0.11 of AUROC and 0.20 of average precision that the model gets only from recognising relatives.

I am reporting the honest number because the whole point of this exercise is judging genes the model has never specifically seen drugged before, and that is the real situation I am actually in with LUAD. It also explains why a prior like FGFR2's is easy to trust (its sibling FGFR1 is drugged), while a prior for a gene with no close drugged relative deserves more caution.

4. What the anchor gene variants taught me

This course asks everyone to check the same four anchor variants regardless of disease, and one result genuinely surprised me. APOE's two famous variants came out backwards from what I expected. I use UniProt numbering, which includes APOE's 18-residue signal peptide, so C130R and R176C correspond to the mature-protein names C112R (ε4) and R158C (ε2).

Gene	Variant	Score	Percentile in protein
TP53	R175H	−5.97	19.3
APOE	C130R (ε4)	+7.21	100.0
APOE	R176C (ε2)	−8.56	11.8
TREM2	R47H	−5.16	47.7

The APOE ε4 variant is the single strongest known genetic risk factor for Alzheimer's and is associated with real elevated disease risk in actual patients. Yet when I scored it using pure evolutionary logic, it came out as one of the most acceptable and unremarkable changes possible anywhere in that protein. Meanwhile ε2, the variant that is actually protective against Alzheimer's, scored as one of the most alarming and evolutionarily rare changes in the entire protein, which is nearly the opposite of what I would guess from the clinical outcome.

I think the explanation is that ε4 is actually the older ancestral version of this gene across primates, so evolution has seen it for a very long time and does not consider it unusual, even though it only becomes a real problem for people living long enough in a modern world for Alzheimer's to show up. This is why a strongly negative score is not the same as “pathogenic”. The score only tells you that a change is rare by evolution's standards. A rare change can be harmless or even protective (like ε2), and a common, “normal-looking” one can still raise disease risk (like ε4).

The other two anchors are a reminder that even real and clinically important mutations do not always show up as the most dramatic outliers. TP53's R175H, a textbook cancer hotspot, scored as more unusual than about 80% of possible changes in TP53 (19th percentile), but not at the extreme. TREM2's R47H, a known Alzheimer's risk variant, sat right in the middle (48th percentile).

There are also things these models cannot see at all. They only look at one protein sequence at a time, so they have no idea what the gene does in a human brain or tumour, which tissue it matters in, how common the variant is in real populations, which other variants a person carries alongside it (ε2 and ε4 are themselves defined by a combination of two positions), or how a variant changes protein behaviour, such as a mutant p53 interfering with the normal copy.

5. What I still don't know, and where I'd look next

None of this tells me whether a drug could physically fit onto any of these proteins, since that requires an actual 3D shape and not just a sequence. It also cannot see anything about how a protein behaves once it is actually inside a cell, attached to its real partners, or chemically modified after being built, which is all real biology that a sequence alone just does not capture. A high prior only means “this looks like proteins we have drugged before”, not “this protein has a usable pocket” or “drugging it would help LUAD patients”.

Two targets for Week 4: FGFR2 and CYP1A1. These two have the strongest and most believable small-molecule priors in my list (p_sm 0.86 and 0.75), and in both cases the evidence comes from a nearly identical drugged relative (FGFR1 at 0.99 for FGFR2, and CYP17A1 at 0.99 for CYP1A1), not a loose match. Both are enzymes with well-known binding sites, a kinase ATP pocket in FGFR2 and a heme-containing active site in CYP1A1, so a structure-based pocket assessment is a fair test of whether the prior holds up. They also cover two different stories: FGFR2 as a classic growth-signalling receptor, and CYP1A1 as the tobacco-carcinogen enzyme. CYP1A1's ranking also stayed stable in my Module 2 sensitivity check using only LUAD-specific genetic evidence, while FGFR2's top position partly depended on cross-cancer genetic evidence. I did not pick SLC12A7, even though its p_sm is also high (0.85), because its closest matches are ion channels from a different protein family, so its prior is less specific.

Summary. Adding a druggability prior kept FGFR2 at the top, moved likely drug-like proteins (SLC12A7, CYP1A1, VWF) up and structural proteins (BRCA2, PALM) down, and showed clearly how location decides modality (VWF → antibody, FGFR2 → small molecule). The family-split validation says the prior is real but modest, and the anchor variants show that evolutionary scores measure how unusual a change is, not how dangerous it is. FGFR2 and CYP1A1 go forward to Week 4 for structure-based pocket assessment.

References

1. Bouaoun, L., Sonkin, D., Ardin, M., Hollstein, M., Byrnes, G., Zavadil, J., & Olivier, M. (2016). TP53 variations in human cancers: New lessons from the IARC TP53 database and genomics data. *Human Mutation*, 37(9), 865–876. https://doi.org/10.1002/humu.23035
2. Pacini, L., Jenks, A. D., Lima, N. C., & Huang, P. H. (2021). Targeting the fibroblast growth factor receptor (FGFR) family in lung cancer. *Cells*, 10(5), 1154. https://doi.org/10.3390/cells10051154
3. Scully, M., et al. (2019). Caplacizumab treatment for acquired thrombotic thrombocytopenic purpura. *New England Journal of Medicine*, 380(4), 335–346. https://doi.org/10.1056/NEJMoa1806311
4. ESM Team (2024). ESM Cambrian: Revealing the mysteries of proteins with unsupervised learning. EvolutionaryScale. https://www.evolutionaryscale.ai/blog/esm-cambrian